Zeta-Law Entropy and Zeta-Inequality Framework

Abstract

This note records the audited core of a zeta-law framework for treating zeta-ratio identities, modular entropy shadows, and applications to inequalities for the Riemann zeta function. The fully verified material in this version consists of the Gibbs-law formulation, free-energy notation, modular and Mellin-Planck definitions, and Euler-score identities. Several stronger application claims are recorded as audit targets.

1 The zeta law

Definition 1 (Riemann zeta probability law)**.** For $\beta > 1$, define the probability law ρ_β on $\mathbb{N}$ by

$$\rho_\beta(n) = \frac{n^{-\beta}}{\zeta(\beta)}, \qquad n \in \mathbb{N}.$$

Equivalently,

$$\rho_\beta(n) = \frac{e^{-\beta E(n)}}{Z(\beta)}, \qquad E(n) = \log n, \qquad Z(\beta) = \zeta(\beta).$$

Definition 2 (Zeta free energy)**.** The zeta free energy is

$$A(\beta) = \log \zeta(\beta), \qquad \beta > 1.$$

Differentiating under the absolutely convergent Dirichlet series gives

$$A'(\beta) = -\mathbb{E}_\beta[\log N], \qquad A''(\beta) = \mathrm{Var}_\beta(\log N).$$

Remark 1. For $r \geq 0$, the zeta law satisfies the moment identity

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta + r)}{\zeta(\beta)}.$$

Thus many zeta-ratio inequalities can be read as moment inequalities under ρ_β.

2 Modular successor entropy

Definition 3 (Residue distribution and successor entropy)**.** For $q \geq 2$, define the residue distribution of the zeta law modulo q by

$$\mu_{q,\beta}(a) = \sum_{\substack{n \geq 1 \\ n \equiv a \pmod q}} \rho_\beta(n), \qquad a \in \mathbb{Z}/q\mathbb{Z}.$$

Writing $a+1$ cyclically in $\mathbb{Z}/q\mathbb{Z}$, define

$$B_q(\beta) = \sum_{a \in \mathbb{Z}/q\mathbb{Z}} \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\mu_{q,\beta}(a+1)}.$$

Problem 1 (Successor entropy resolution)**.** Audit the claimed identity

$$\Sigma(\beta) = \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta+k),$$

together with the modular variational formula

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \to \infty} B_q(\beta).$$

The intended proof expands $\log(1+1/n)$, uses the log-sum inequality, and lets large moduli isolate finite initial intervals.

Problem 2 (Prime-modulus Dirichlet L-resolution)**.** For prime p, let χ range over Dirichlet characters modulo p, and set

$$S_a(p,\beta) = \sum_{\chi \bmod p} \chi(a) L(\beta, \chi).$$

Audit the claimed resolution

$$\mu_{p,\beta}(a) = \frac{S_a(p,\beta)}{(p-1)\zeta(\beta)} \quad (1 \leq a \leq p-1), \qquad \mu_{p,\beta}(0) = p^{-\beta},$$

and the resulting prime-modulus formula for the successor entropy.

3 Euler-score identities

Proposition 1 (Euler-score decomposition)**.** *For every* $\beta > 1$*,*

$$-\big(\log \zeta\big)'(\beta) = \mathbb{E}_\beta[\log N] = \sum_{d=1}^{\infty} \Lambda(d)\mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}.$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}.$$

Proof. Use the von Mangoldt identity

$$\log n = \sum_{d|n} \Lambda(d).$$

Then

$$\mathbb{E}_\beta[\log N] = \sum_{n\geq 1} \rho_\beta(n) \sum_{d|n} \Lambda(d) = \sum_{d\geq 1} \Lambda(d)\, \mathrm{P}_\beta(d \mid N).$$

For the zeta law,

$$\mathrm{P}_\beta(d \mid N) = \frac{1}{\zeta(\beta)} \sum_{m\geq 1} (dm)^{-\beta} = d^{-\beta}.$$

The identity with $-(\log \zeta)'(\beta)$ follows from the free-energy derivative. □

Proposition 2 (Finite Euler-score identity)**.** *For every* $A \subseteq \{1, \ldots, N\}$*,*

$$\sum_{n\in A} \log n = \sum_{d\leq N} \Lambda(d) \sum_{m\leq N/d} \mathbf{1}_A(md).$$

Proof. Reverse the order of summation:

$$\sum_{d\leq N} \Lambda(d) \sum_{m\leq N/d} \mathbf{1}_A(md) = \sum_{n\in A} \sum_{d|n} \Lambda(d) = \sum_{n\in A} \log n.$$

□

4 Mellin-Planck partition function

Definition 4 (Mellin-Planck partition function)**.** For $s > 1$, define

$$M(s) = \Gamma(s)\zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1}\, dt.$$

The normalized law associated with M is

$$\nu_s(dt) = \frac{t^{s-1}}{(e^t - 1)M(s)}\, dt.$$

Remark 2. The free-energy identity

$$\frac{d^2}{ds^2} \log M(s) = \mathrm{Var}_{\nu_s}(\log t) \geq 0$$

shows that M is log-convex on $(1, \infty)$.

5 Application audit targets

Problem 3 (Nantomah positivity)**.** Audit the claimed proof that, for every $n \in \mathbb{N}$,

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0.$$

The imported route rewrites the expression using moments of $X(m) = 1/m$ under ρ_s, then splits the result into a positive linear part and a bounded quadratic correction.

Problem 4 (Alzer-Kwong reciprocal-zeta curvature)**.** Let $F(x) = 1/\zeta(x)$. Audit the claimed sign pattern

$$F''(x) > 0 \quad \text{on } (-4n, -4n+2), \qquad F''(x) < 0 \quad \text{on } (-4n-2, -4n)$$

for every integer $n \geq 1$. The imported route uses the zeta functional equation and a positive-axis curvature bound for $(\log \zeta)''(s)$.

Problem 5 (Sroysang generalized Holder inequality)**.** Let $m \geq 2$, $r \geq 1$, $x_1, \ldots, x_m > 1$, and $p_1, \ldots, p_m > 1$ satisfy

$$\sum_{i=1}^{m} \frac{1}{p_i} = \frac{1}{r}.$$

With

$$T = r \sum_{i=1}^{m} \frac{x_i}{p_i},$$

audit the claimed inequality

$$\zeta(T) \leq \frac{\prod_{i=1}^{m} (\Gamma(x_i)\zeta(x_i))^{r/p_i}}{\Gamma(T)}.$$

The imported route applies generalized Holder to

$$f_i(t) = \left(\frac{t^{x_i - 1}}{e^t - 1} \right)^{1/p_i}.$$

6 Program

The framework currently separates four proof layers:

$$\frac{\zeta(s+r)}{\zeta(s)} = \mathbb{E}_s[N^{-r}], \qquad (\log \zeta)''(s) = \mathrm{Var}_s(\log N),$$

$$M(s) = \Gamma(s)\zeta(s), \qquad (\log M)''(s) = \mathrm{Var}_{\nu_s}(\log t),$$

and

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta).$$

The next mathematical task is to audit the imported application proofs and promote each application from problem status to theorem status only after the proof is self-contained and all endpoint, exponent, and extraction-sensitive formulas have been checked.